

$$\begin{aligned}
 1 \quad (a) \quad (i) \quad V &= \pi r^2 h \\
 &= \pi \left(\frac{0.04261}{2} \right)^2 (0.126) \\
 &= 1.797 \times 10^{-4} m^3
 \end{aligned}$$

$$\begin{aligned}
 \frac{\Delta V}{V} &= \pm \left(2 \frac{\Delta r}{r} + \frac{\Delta h}{h} \right) \\
 &= 2 \frac{0.02}{42.61} + \frac{0.2}{12.6} \\
 &= 1.68\%
 \end{aligned}$$

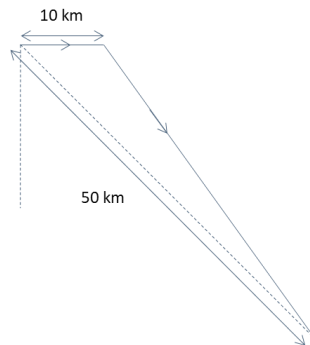
$$\Delta V = 3 \times 10^{-6}$$

$$V = (180 \pm 3) \times 10^{-6} m^3 = 180 \pm 3 cm^3$$

(ii) Since the absolute/percentage uncertainty of the measuring cylinder is smaller than that of (a), the method is more accurate

(iii) take repeated measurement and find average

(b)



20.5 km h⁻¹ to 23.0 km h⁻¹ and 53° to 57°